Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 0 to Draft 1

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Scope

Draft 0 of the companion paper set up the rank-two wild problem and proved positivity of the character transform $L_D\mathbf{1}$ on all covered quadratic tori, leaving Target A (is $L_D\mathbf{1}$ an irreducible genuine character of the 3-fold cover of $D^\times$ over $F = \mathbb{Q}_3(\zeta_3)$?) open. Draft 1 resolves Target A positively, settles Target B for level-zero representations and their norm twists, and records a program for positive level. Sections 2–6 of Draft 0 are retained verbatim; new material is Sections 7–12, the interface section, and two appendices. The status labels [P]/[E]/[O]/[C] are kept; one new label [P^c] ("proved modulo a finite exact computation whose script is supplied") is introduced and explained in the status table.

Mathematical additions

1. **Depth structure of K^1 (Section 7).** Cubing $K^k \to K^{k+4}$ is bijective for $k \geq 3$; the cube map on $Q = K^1F^\times/F^\times$ is injective; cubes of depth-one elements have depth exactly 3. Hence $\mathrm{Good}(Q) = G_3 \sqcup Q_6$ with G_3 the set of cube classes of depth-one elements. In the coordinates $V = Q_3/Q_6 \simeq \mathbb{F}_3^5$ the image $\bar{G}_3$ has 72 elements lying on 36 lines through 0 and is given by an explicit formula (Proposition `prop:G3explicit`); G_3 is Q_6-invariant. Consequently $\Phi = L_D\mathbf{1}|_{\tilde{U}_1}$ equals $27\zeta\tilde{\theta}$ on star lifts over U_6, $9\zeta\tilde{\theta}$ on star lifts over $F^\times G_3$, and 0 elsewhere (Lemma `lem:Phi`).

2. **Brauer reduction (Section 8, Theorem `thm:reduction`).** Target A is equivalent to Φ being a virtual character of $\tilde{U}_1$, and then automatically an irreducible character of the finite quotient $\Gamma_0 = \tilde{U}_1/N$. Ingredients: the finite quotient $\Gamma = \tilde{H}/N$ with $\Gamma/\Gamma_0 \simeq D_8$; the 2-group lemma (the star section is the unique complement to μ_3 in $\Gamma_{P_0} \simeq \mu_3 \times D_8$); the centralizer lemma, proved via the σ-decomposition of pro-3 groups with an involution; and Weyl-integration computations of $\langle \Phi|_{\tilde{U}_m}, \Phi|_{\tilde{U}_m} \rangle = 1, 27, 27, 729$ for $m = 1, 3, 4, 6$, which give $\langle \Phi, \Phi \rangle = 1$. A remark records that the choice of the cube-root character θ_0 is immaterial.

3. **Hypothesis (H1) and the finite design (Section 9).** (H1) states that ψ_6 ("$\zeta\tilde{\theta}(\tilde{z})$ on star lifts") is a character of $\tilde{U}_6$. Its consequences are worked out: the commutator pairing $\kappa_{1,5}(a, b) = \zeta^{\pm \mathrm{Tr}(a\bar{b})}$, the vanishing of the commutator form ω on $\tilde{V} = \tilde{U}_3/N_6$, and the factorization of the star section through $\bar{G}_3$. Under (H1) the restriction of any candidate τ to $\tilde{V}$ is multiplicity free with 27 characters Λ permuted simply transitively by the Heisenberg group Q_1/Q_3. The resulting combinatorial problem (an odd, D_8-invariant defect function e on the 36 lines satisfying Fourier

conditions $(9,9,9)/(15,6,6)$ and a Heisenberg cocycle identity) has exactly two solutions $\pm e_*$, and $e=0$ is excluded by $\kappa_{1,5}(1,1)\neq 1$ (Propositions `prop:forces`, `prop:design`). Since $0\in\Lambda$, Theorem `thm:H1equiv` gives the closed form

$$\tau|_{\tilde{U}_1}=\mathrm{Ind}_{\tilde{U}_3}^{\tilde{U}_1}\lambda_0,\qquad \lambda_0(y)=\Big(\prod_{d\in D_8}\tilde{d}\,y\,\tilde{d}^{-1}\Big)^{-1},$$

with the product evaluated through ψ_6.

4. **Explicit cover model and proof of (H1) (Section 10).** Using $\iota(x+yj)=\left(\begin{smallmatrix} x & \varpi y\\ \sigma(y) & \sigma(x)\end{smallmatrix}\right)$ and a Kubota–Kazhdan–Patterson cocycle on $\mathrm{GL}_2(E_0)$ twisted by $(\det,\det)_{E_0}^{c_v}$, with the cubic Hilbert symbols of F and E_0 determined as the unique (up to sign) alternating forms killed by the Steinberg relations, the cube identity $(w,0)^3=(w^3,0)$ is proved by hand for $w=x+yj$, $y\neq 0$, $\mathrm{Tr}\,x\neq 0$, $(\mathrm{Tr}\,x)^2-\mathrm{Nrd}\,w\neq 0$ (Proposition `prop:cubeidentity`). This yields the star section on K^6 in closed form and reduces (H1) to the identity $\kappa(XX')=\kappa(X)+\kappa(X')+\beta(X,X')$ for an explicit $\kappa(X)=(z_X,y_X)_{E_0}$ and $\beta(X,X')=-h_E(r,t)-h_E(r,n)$. Cases (i)–(iii) of this identity are proved by hand; the boundary cases (iv)–(v) were verified on a stratified exhaustive-by-class set of 7600 exact instances (Proposition `prop:H1`, Remark `rem:H1-status`). A second, independent route (Lazard splitting over the uniform subgroup K^4; the space of invariant alternating forms on K^4/K^8 vanishing on torus subspaces is zero, and the equivariant defect on Q_6/Q_8 is zero modulo linear) is recorded.

5. **Main theorem and corollary (Section 11).** Theorem `thm:main-targetA`: Target A holds; $L_D\mathbf{1}$ is the character of a single 27-dimensional genuine irreducible representation τ of $\tilde{H}$, with $\tau|_{\tilde{U}_1}$ as above. The end-to-end check (induced character of λ_0 agrees with Φ at every tested point, including the phases $9\zeta^m$ on $F^\times G_3$ and the zeros on non-good elements) is recorded. Corollary `cor:levelzero`: the five level-zero ρ (irreducible representations of D_8) and their twists $\rho\otimes\chi\circ\mathrm{Nrd}$ with $\chi|_{\mu_3}=1$ also have $L_D\Theta_\rho$ an irreducible character.

6. **Targets B and C (Section 12).** Target B at positive level and Target C remain [O]/[C]; the section lists what the present method needs (an analogue of the depth lemma and design search for each depth, and the Brauer reduction already applies verbatim).

Status changes

Item	Draft 0	Draft 1
Positivity on quadratic tori ($r=2$)	[P]	[P]
Brauer reduction to $\tilde{U}_1$	—	[P]
Depth lemma, G_3, Φ formula	—	[P]
Finite design uniqueness	—	[P^c]
Cube identity, star section on K^6	—	[P]
(H1)	—	[P^c] (cases (i)–(iii) [P]; (iv)–(v) computational)
Target A	[O]	[P^c]
Target B, level zero and norm twists	[O]	[P^c]
Target B, positive level	[O]	[O]
Target C	[C]	[C]

The [P^c] label is deliberately not collapsed to [P]: the finite computations are exact (integer and $\mathbb{Q}(\zeta_{12})$ arithmetic, no floating point) and their scripts are supplied, but they have not been replaced by a written proof. The one non-exhaustive step is the boundary case verification in Proposition `prop:H1`; turning it into a proof (or an exhaustive enumeration over Q/Q_9) is the first item of future work.

Computational record

All scripts are in `part-IV/scripts/` and listed in Appendix B of Draft 1. `quat.py` implements exact arithmetic in $\mathbb{Z}[i][\varpi]$ and O_D; `cover.py` implements the cover model over $\mathbb{Q}(\zeta_{12})$ with the Hilbert symbols solved from the Steinberg relations. Steps 1–6 (depth lemma data, Q/Q_7, actions on V, design search, Heisenberg extension), step 7 (symbols), steps 8–10 ((H1) tests, defects, $\omega = 0$, $\kappa_{1,5}$), and step 11 (end-to-end induced character) are each independent and reproducible in minutes. Appendix A of Draft 1 tabulates the action matrices, the 36 lines with e_*, the set Λ, the Heisenberg cocycle data, and the two Hilbert pairings.

Editorial changes

New abstract; status table extended with the [P^c] label; introduction paragraphs on the method (Brauer reduction, finite design, explicit cocycle) and on what remains; interface section recording which Part I results are used (Prop. 2.3, Prop. 3.1, Prop. 3.5, Thm. 3.9, Remark 8.1) and which Draft 0 statements are superseded; bibliography extended with Kazhdan–Patterson, Lazard/Dixon–du Sautoy–Mann–Segal, Moore, and Isaacs. American English spelling throughout. No numbered statement of Draft 0 was renumbered.

Repository consistency

Part III, Draft 9, and the running-example note refer to the wild-place integrality question as open; when they are next revised they should cite Draft 1, Theorem `thm:main-targetA`, with its [P^c] qualification. The scripts directory should be committed alongside the TeX source so the [P^c] claims are checkable.

Verification performed

Draft 1 was compiled twice with `pdflatex` (25 pages); all internal references resolve. Every numerical claim in Sections 7–11 and Appendix A was regenerated from the scripts during drafting. As with any research manuscript, external specialist refereeing is advisable, in particular of the centralizer lemma and of the hand-proved cases of Proposition `prop:H1`.